ANALYZING BIASES IN LOAN APPROVAL MACHINE LEARNING MODELS TRAINED ON HMDA DATA

Anthony Baez, Rodrigo Garza, Dami Thomas, & Waris Tuchinda
Massachusetts Institute of Technology
{acbaez, rgg1, dvthomas, willytcd}@mit.edu

ABSTRACT

Machine learning models are increasingly being used for home mortgage approvals, raising concerns about bias amplification from historical data. In this paper, we examine biases in loan approval models trained on the Home Mortgage Disclosure Act (HMDA) dataset, comparing performance across multiple machine learning models. We evaluate model accuracy, F1 scores, and fairness metrics when including or excluding sensitive features like race and location. Results show that while XGBoost achieves the highest predictive accuracy, it also exhibits significant bias. Logistic Regression, by contrast, balances fairness and performance more effectively. We then discuss implications for mitigating biases in financial AI systems.

1 INTRODUCTION

1.1 PROBLEM DEFINITION

Biases in financial systems have long been a driver of economic inequality Ruggeri et al. (2023). While the influence is multi-faceted, biases have a significant impact on influencing individuals' access to financial credit HAI (2023). The issue of biases in the financial sector may only be magnified as financial institutions adopt machine learning models to assist and streamline decision-making processes. While promising efficiency and accuracy, these automated models come with the risk of amplifying societal biases in the financial system. If trained on historical training data that are products of discriminatory practices, these machine learning models could learn the biases and, as a result, make decisions that further exacerbate biases and inequalities in our society. These decisions would cause discriminated groups to perform worse financially and create more biased data points, resulting in a self-perpetuating cycle of discrimination PLC (2019). To break this cycle of discrimination, more research has to be done to understand how machine learning models might develop biases with the inclusion or exclusion of sensitive features such as race, gender, or residential location. We must also understand how these features relate to predictive accuracy.

In this research, we compare the performance and bias of various machine learning models trained to decide whether or not to approve an application for a home mortgage while both including and excluding sensitive potentially biasing features, of which we focus on race. We chose to focus on mortgage approvals as they dramatically influence both individual economic opportunities and broader wealth-building in society. Disparities in mortgage approvals not only disadvantage individuals but also reinforce systemic inequalities Bartlett et al. (2022). Identifying the mechanisms of bias formation in how these models deal with loan approval is crucial for fostering equity as we brace for the rising role of machine learning-based decision-making models in the financial sector.

2 RELATED WORKS

Biases in automated loan-credit approval systems are an important subject, so there has been much prior research. Starting from identifying intrinsic bias in historical loan-approval statistics, Zou & am nuai (2023) identified the presence of ethnic bias in historical mortgage approvals. Their research shows that Black applicants face significantly higher denial rates on requests than White applicants (controlled for similar financial profits). Their work also touched on the potential of AI systems to

replicate these biases if trained on said data. Next, it is important to understand that there seems to be a fundamental trade-off between fairness achieved for models and other objectives such as predictive accuracy, as explored by Lee & Floridi (2020). The trade-off between fairness and other objectives warrants research that goes into understanding how to de-bias these models. Tenev (2024) showed how the inclusion or exclusion of sensitive features affects the XGBoost bias score and accuracy. After understanding the underlying bias in training data, the trade-off between bias and other metrics such as predictive accuracy, and how to de-bias techniques such as inclusion/exclusion of features, the next step is to understand the long-term implications of attempting to make these automated systems more fair as explored Kumar et al. (2022). Zhang and Hosoi argue that if left without interventions, the automated mortgage lending process could greatly exacerbate existing disparities even under unbiased policies and regulations from the government Zhang & Hosoi (2024). In the hope of offering a different research angle to these related works, our research goal is to cover all of the exploration steps from uncovering underlying bias in our training data, all the way to selecting and training machine learning models and learning about how the models pick up bias and how that relates to predictive accuracy.

3 Approach

3.1 Research Objectives

Our main research question is whether we can develop machine learning models that are able to accurately approve mortgage loans based on the decisions of past approvals, as well as how these trained models behave as they relate to perpetuating or mitigating existing biases in lending practices.

Our research has several main objectives. The first is to quantify and explore the relationship between model accuracy and bias for each of our potentially biasing features. We aim to find out what trade-offs exist between accuracy and bias and try to determine whether high accuracy on our approval task necessitates high bias, indicating that historical loan approvals are themselves biased.

The second is to analyze the relationship between certain biasing features and whether the inclusion or exclusion of them affects the predictions of the models. If adding or omitting certain sensitive features, like race, sex, or location, changes predictions or the model accuracy significantly, then we conclude that the model is weighting those features higher when making predictions. This could indicate that the model, and therefore the original home loan approval process, were biased.

The third is simply seeing how bias and accuracy trade-offs differ depending on the model architecture. For example, are simpler, more traditional statistical and machine learning models more effective than a neural network? If so, how do their biases compare?

3.2 Dataset Overview

Our project makes use of the Home Mortgage Disclosure Act (HMDA) dataset. In particular, the HMDA dataset provides records from the last two decades, with each year containing more than 10 million records. Due to computational limitations, we randomly sampled 500,000 records of the 14,285,496 available in the 2017 dataset to analyze, since 2017 is the latest year to have complete data. A size of 500,000 was chosen as it was enough samples to provide meaningful statistical results while still capturing meaningful relationships in the data for the training of our models. Because of the size of the dataset ($\sim$11 GB), the dataset was loaded using chunks of 100,000 records each. Random sampling was then conducted on each chunk.

3.3 Exploratory Data Analysis

Before model training, it was crucial to explore the data and the relationship between its features using the Pandas Python package. We first made a heatmap showing the approval rates for different races given different income brackets using the data directly from the dataset (Figure 1). This shows us what disparities already exist in the data. In this figure, the darker the square, the higher the approval rate. We can see that approval rating intuitively increases with income, but we can also see that across races, particularly in lower incomes, there is a significant disparity in approval rates.

Specifically, even at the same low income, Black, American Indian, and Pacific Islander groups have more than a 10% lower approval rating than White and Asian.

Another important plot was the feature correlation matrix, which shows us which features are most correlated with loan approvals, which can be seen by looking at the bottom row of the matrix (Figure 2). We see that $minority_population$, which is the fraction of a given county that is non-white, has the lowest value. This is insightful because the loan approval process cannot use race data, but they can use these aggregate location features that act as proxies for race.



Figure 1: Correlation matrix showing the effect of race and income on approval rate in HMDA dataset

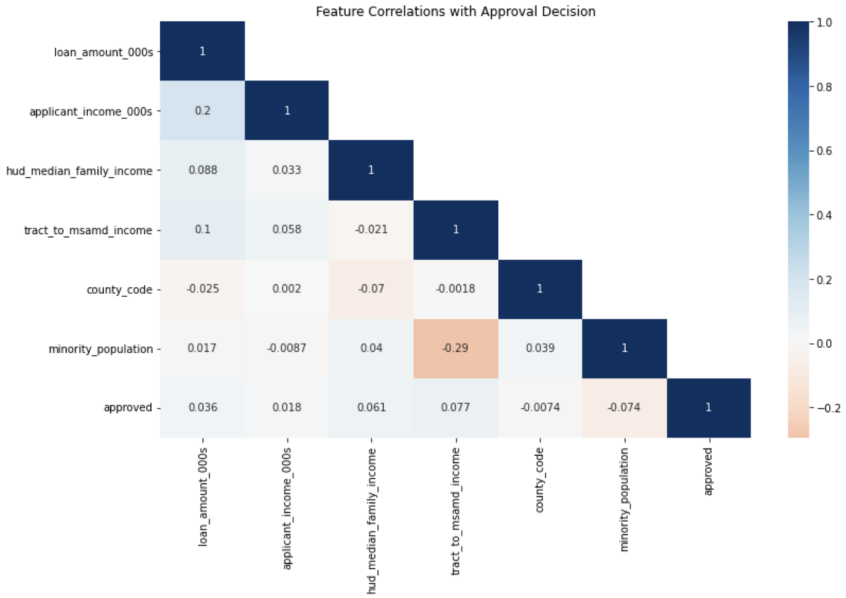


Figure 2: Correlation matrix showing the effect of features on approval rate in HMDA dataset

3.4 Feature Selection

We divided our dataset into three sets of features. The first set was our baseline features, which contained $loan_amount_000s$, $applicant_income_000s$, $hud_median_family_income$,

and $tract_to_msamd_income$. We then had a location set that contained $state_code$, $county_code$, and $minority_population$. Finally, we made a race set that contained the feature $applicant_race_name$.

3.5 Data Pre-processing

For each feature x_i, we normalized each feature in the traditional models because values within each feature varied greatly. We normalized according to the equation $x_{norm} = \frac{x_i - \mu_i}{\sigma_i}$, where μ_i and σ_i represent feature-specific mean and standard deviation. In the neural network, we only divided each feature by its mean, $x_{norm} = \frac{x_i}{\mu_i i}$, as we found that normalization harmed performance. For rows with any missing values, we completely removed the row from the dataset, leaving us with $\sim$100,000 fewer data points.

We also needed to encode the race data into numerical data, as they were recorded as strings such as 'white' in the HMDA dataset. In the traditional models, we used the LabelEncoder package in scikit-learn to convert then record embeddings. For the neural network, we converted the race categories into one-hot embeddings.

3.6 Model Descriptions

We used three traditional machine learning models and one deep neural network model:

Random Forest is a supervised machine learning model that uses an ensemble of multiple different models of binary decision trees to make predictions. It then takes the average of the outputs of these models to be the final output. This is a powerful model because the combination of multiple smaller models improves generalization and reduces variances between different outputs on the same data.

XGBoost is similar to Random Forest, as it contains multiple models of binary decision trees that are compiled for the overall output. However, it applies gradient descent within each binary decision tree to improve performance. This makes it a more powerful model than Random Forest.

Logistic Regression is a supervised learning algorithm used for binary classification tasks. It models the relationship analytically between input features and the probability of a specific outcome using the logistic (sigmoid) function, which maps real numerical values to values between 0 and 1. The decision is determined by applying a decision threshold (usually 0.5).

Deep Neural Networks are a type of artificial neural network with multiple layers between the input and output layers. They use hidden layers composed of neurons that apply learned weights and non-linear activation functions to capture complex patterns in data. Backpropagation and gradient descent are then used to learn from the data. The final activation layer can be made a sigmoid function to make the model into a binary classifier. For this specific model, we use four hidden layers with the largest layer having 1024 neurons.

3.7 Model and Training Setup

We used a randomized 80/20 train-test split for all model training. For the Random Forest and XGBoost, 100 binary classification trees were used in the ensemble. For the neural network, we used a batch size of 100, binary cross-entropy loss, and ADAM optimizer. Our feature groups led to us using three datasets: $baseline$, $baseline + location$, and $baseline + location + race$.

4 Evaluation

4.1 Metrics

Our final dataset included 500,000 mortgage applications with 271,073 approved loans for a baseline approval rate of 54% approval rate. We use this dataset with our four models: Random Forest, XGBoost, Logistic Regression, and Deep Neural Network. The evaluation metrics were calculated using the model's performance on the test set and are as follows:

Accuracy is the proportion of correct predictions of whether or not an applicant was approved for a loan out of the total loan approval predictions.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Precision is the proportion of correctly predicted positive instances out of all **positive predicted** instances. This metric is used to evaluate the reliability of the tested model's classification for the positive class.

$$\text{Precision} = \frac{\text{Correct Positive Predictions}}{\text{Total Predicted Positives}}$$

Recall is the proportion of correctly predicted positive instances out of all **actual positive** instances. This reflects the model's sensitivity to the positive class.

$$\text{Recall} = \frac{\text{Correct Positive Predictions}}{\text{Total Actual Positives}}$$

F1 Score is meant to penalize extreme values and provide a measure of the model's effectiveness in identifying the positive class. It combines precision and recall into one metric.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Each metric essentially compares the tested model's predictive performance to the actual results of the HMDA dataset, giving us a more holistic overview of the results. With these metrics, we chose **Accuracy** and **F1 Score** as the metrics to compare across all of our models. Together, they provide a good overview of performance while offering a deeper understanding of how well the models perform with complex datasets such as the HMDA dataset.

4.2 Results

According to Table 1 and Figure 3, which show the accuracy of all four models after training, we can see that XGBoost consistently performed best out of the four models. This is true regardless of the feature set they were trained on. This shows that XGBoost was the most effective at predicting whether an applicant was approved for a loan. Overall, the results also show that the three traditional machine learning models, Random Forest, XGBoost, and Logistic Regression, have similar performance in accuracy of around 80%. Lastly, the neural network model has a noticeably lower accuracy of around 61%. This was surprising, considering the usefulness and expression ability of neural networks. We hypothesize that the features in the dataset were too noisy and did not allow the neural network to learn the non-linear relationships between features. The traditional methods, on the other hand, learn simpler relationships that might be in the dataset. Additionally, since we made the architecture from scratch, we did not have enough time to try all possible combinations of a number of layers, neurons per layer, and activation functions that might have been suitable for the task.

These trends also hold for F1, which can be seen in Table 2 and Figure 4. XGBoost yields the best F1 scores, at 89%, and neural network has the worst, at 74%. This shows that XGBoost had the best balance between precision and recall, that is predicting positive cases correctly and being able to discriminate against negative cases well.

4.3 Performance with and without Race

Another surprising result was that the accuracy across all models did not differ by more than 2% across all models. This leads to the hypothesis that racial data is implicitly present in the other financial features. As seen in Figure 2, there are high correlations between some of the financial features and other features that can be seen as a proxy for race, such as location. Specifically, we see that the most correlated two features, which are negatively correlated, are the minority population

Model	Baseline	Baseline + Location	Baseline + Location + Race
Random Forest	0.797	0.805	0.808
Logistic Regression	0.797	0.797	0.797
XGBoost	0.808	0.809	0.811
Neural Network	0.613	0.602	0.598

Table 1: Accuracy Values Across Models and Feature Sets

Model	Baseline	Baseline + Location	Baseline + Location + Race
Random Forest	0.883	0.888	0.889
Logistic Regression	0.886	0.887	0.887
XGBoost	0.890	0.891	0.891
Neural Network	0.741	0.736	0.748

Table 2: F1 Score Values Across Models and Feature Sets

and local income. This redundancy in information most likely led to the lack of improvement in accuracy and F1 when you add explicit race data into the models.

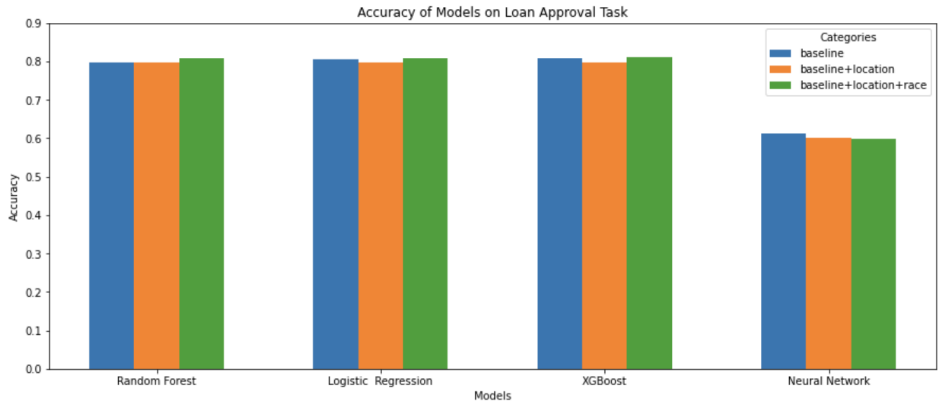


Figure 3: Accuracy for all Trained Models

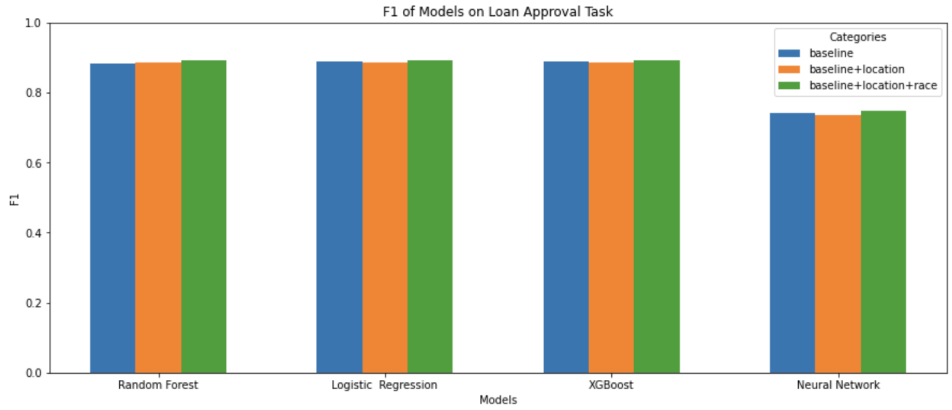


Figure 4: F1 for all Trained Models

4.4 Feature Impact of XGBoost

Seeing that XGBoost performs best for both accuracy and F1 scores, we take a closer look at what contributed to its performance. We can see the relative importance of the different features in the predictions of XGBoost in Figure 5 and Figure 6. We see that when XGBoost was trained on the baseline+location dataset in Figure 5, the most important feature was the financial background of the applicant. This is to be expected, as the manual approval process was based on these features. However, looking at XGBoost when trained on the baseline+location+race dataset, we can see that the race name of the applicant became the most important feature. This shows us that even though the mortgage approval process is race-blind, a model that has race uses it the most to make accurate predictions on mortgage approval. This demonstrates that bias pervades XGBoost when trained on our dataset.

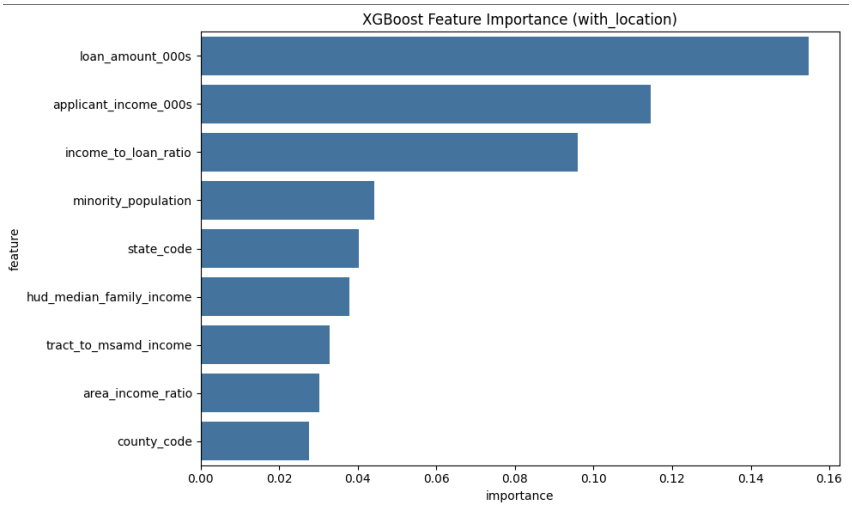


Figure 5: Feature Importance in XGBoost with baseline+location

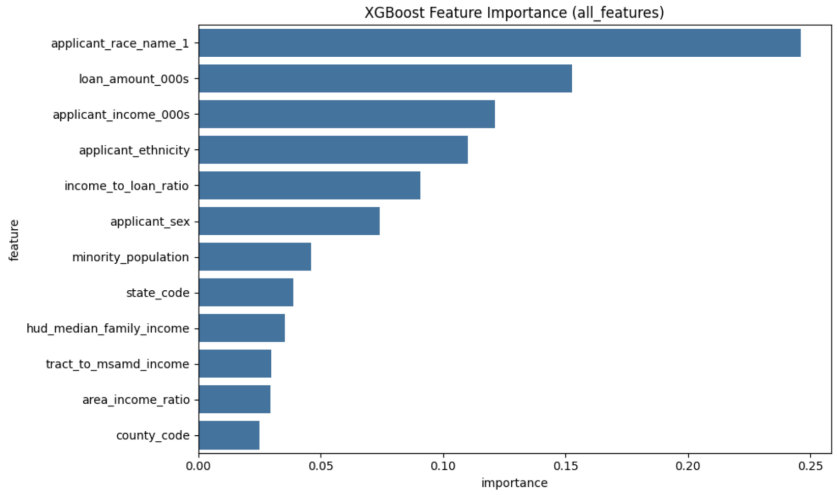


Figure 6: Feature Importance in XGBoost with baseline+location+race

4.5 Disparities Across Races

We see now that XGBoost performs best, in regards to F1 and accuracy, but is also sensitive to using racial features in its prediction. This brings forth the question of whether this accuracy equates to most fair as well. Figure 7 contains the differences in approval rate for each race for each model

and the ground truth dataset, which is referred to as 'historical'. XGBoost and Random Forest have similar disparity rates to the dataset for most races except for Pacific Islanders. Logistic Regression, on the other hand, has by far the least disparities in approval rates, while still having similar accuracy to XGBoost. This suggests that it is actually the least biased model. This trade-off between accuracy and fairness opens a possible area of further research in determining a balance between the two traits for any AI system deployed within a human-interacting setting. We were not able to include the neural network model due to its architecture and time constraints.

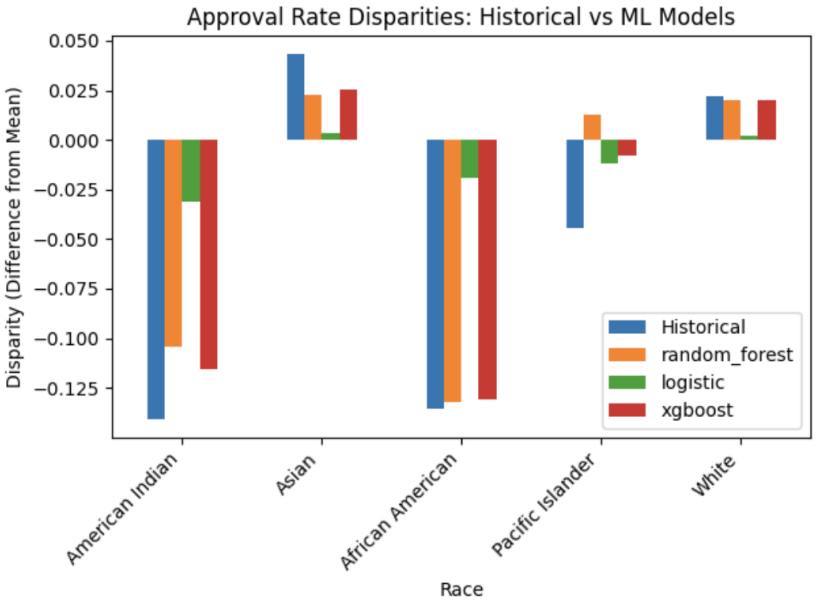


Figure 7: Disparities in Approval Rate across Race for each model (Dataset is baseline+location+rate)

4.6 Limitations

As stated before, pre-existing biases on racial identity were already seen in the HMDA dataset. Considering that this dataset is our baseline standard, this forms a significant limitation on how unbiased we can make our model. One possible alternative to this is using a dataset from a region, state, or other country that does not have a history of housing discrimination like the United States on a national level.

5 Conclusion

From our work thus far, we can conclude that our machine learning models are indeed prone to picking up biases from the datasets they are trained on, with some models being more susceptible than others. We looked at four machine learning models and compared their accuracy and F1, finding that XGBoost performs marginally better than Random Forest and Logistic Regression, and the neural network model performs significantly worse. We also found that significant correlations exist between financial features, such as income, and other features such as location and race, complicating any attempt to make a truly unbiased dataset and model. We also saw that this results in XGBoost using race to make its predictions. Despite these potentially disappointing results, Logistic Regression resulted in significantly less disparity across races, yielding a promising architecture for the further development of unbiased home mortgage approval models.

For manually approved loans, it is expected that the approvals are made from an objective standpoint that is fair to all applicants involved. So, if we hold ourselves to these morally grounded standards, should we not also hold the ML systems we train and deploy to these same standards? With many

manual processes now becoming automated and replacing manual operations, it is important to ensure that efficiency and ease do not sacrifice fairness and societal standards.

Overall, these findings provide strong evidence that current machine learning approaches to mortgage lending decisions unintentionally perpetuate historical lending disparities. The consistency of these patterns across different model types, combined with the strong correlations between demographic factors and approval decisions, points to a fundamental challenge in creating truly fair lending algorithms. This raises important questions about the use of artificial intelligence in financial decision-making, considering that errors made by these systems aren't merely numbers but represent people whose lives are directly affected. Before the deployment of AI systems in financial systems becomes standard, researchers and stakeholders should explore the decision-making and reasoning behind AI systems. This project provides a solid baseline for understanding how models perform given certain socioeconomic and demographic features and for identifying ways to counteract such biases from forming.

However, further work can build upon this idea. One possibility is training a set of ML models, perhaps the same ones used in this project, on HMDA datasets along with features related to an applicant's racial and socioeconomic background and testing to see which applicants the models predict will default on a loan or not. This would be a great opportunity to explore objectiveness in model behavior by examining how the tested model determines an applicant's likelihood of repaying their loans or whether the model relies on biased trends.

Author Contributions

Each member of our team was involved in all aspects of the research project, but each had their own specific focus. Anthony looked into which machine learning models would be best to compare and built the neural network model. He also investigated current literature to identify what areas of our domain had been explored and what could be further developed. Rodrigo handled dataset collection and processing to prepare it for input into our models. Rodrigo also addressed ethical concerns related to home mortgage approval. Dami and Waris took the lead on model training and evaluation to assess performance and potential biases. Dami researched what current institutions use to determine mortgage approval, while Waris focused on measuring actual bias and comparing it across models. We all wrote the research paper together. We are all undergraduate students.

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